Change Detection in SAR Images via Ratio-Based Gaussian Kernel and Nonlocal Theory

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***Abstract*— Compared with the synthetic aperture radar (SAR) image processing theory based on local neighborhood, the nonlocal theory is not limited to a local neighborhood of an image and has great potential in change detection of SAR images. In this study, an approach using ratio-based nonlocal information (RNLI) is proposed for change detection in multitemporal SAR images. First, the RNLI is extracted from a spatial–temporal nonlocal neighborhood where the similarity of two pixels in the nonlocal neighborhood is well characterized by the proposed ratio-based Gaussian kernel function. The parameters of RNLI: noise level and matching window size are adaptively determined to avoid the uncertainty of the change detection result caused by user experience. Second, the difference image is generated by using the RNLI and the ratio operator. Finally, the change map is obtained by segmenting the difference image with a threshold. Experiments conducted on two real datasets and two simulated datasets showed that the proposed method performed better than the other advanced change detection methods, which can better retain the edge information of the changed area while reducing the overall error of the change detection results.**

***Index Terms*— Change detection, nonlocal information, ratio-based Gaussian kernel, spatial–temporal, synthetic aperture radar (SAR).**

I. Introduction

CHANGE detection techniques can be applied to detect the land cover changes by using multitemporal remote sensing images covering the same geographical area [1], which have been widely used in many fields such as urban studies [2], map updating [3], agricultural surveys [4], forest monitoring [5], and disaster assessment [6]. Moreover, such techniques have great potential in dynamic monitoring for the national geographical state of China [7], [8].

Compared with change detection using optical remote sensing images, the accuracy of change detection using synthetic aperture radar (SAR) images is mainly affected by their inherent speckle noise. However, the ability of SAR satellites to obtain ground information at night and in complex weather conditions has attracted scholars from all over the world to study SAR image processing and application techniques. In recent years, with the launch of more SAR satellites and the advancement of SAR image processing technology, change detection in SAR images has attracted great attention from scholars [9]–[12].

Change detection in remote sensing images usually includes the following steps: 1) image preprocessing, 2) comparing multitemporal remote images and generating a difference image, and 3) analyzing the difference image. This study focuses on step 2, which is a key step in change detection of remote sensing images [13]. Early studies over this issue focused on pixel-based change detection methods. Among them, the ratio method [14] and the subtraction method [15] were the two most classic methods for change detection in SAR images. The ratio method was more widely used than the subtraction method in change detection of SAR images as the multiplicative noise model is most suitable for processing SAR coherent image information. Log ratio (LR) was the most popular pixel-based method for change detection of SAR images [12], [16], [17] because the logarithmic transformation used by LR can convert the multiplicative speckle noise of SAR images into additive noise. The pixel-based change detection method is an ideal choice if the SAR images are not disturbed by noise. However, when acquiring SAR images, SAR satellites first actively emit microwaves, and then receive reflected echoes from ground objects. This special imaging process makes multiplicative speckle noise an inherent feature of SAR images [18], [19]. In order to reduce the negative impact of speckle noise on change detection of SAR images, the mean ratio (MR) method uses the local mean of the center pixel to replace its intensity value, which performs better than the classic ratio method [20]. On the basis of the classic local mean, Buades *et al.* [21] proposed a nonlocal mean (NLM) method, in which the similarity of neighboring pixels was considered and the spatial neighborhood information was used more reasonably and effectively. This NLM method has been used in the fields such as imaging recovery [22], nondestructive testing [23], medical image processing [24], [25], super

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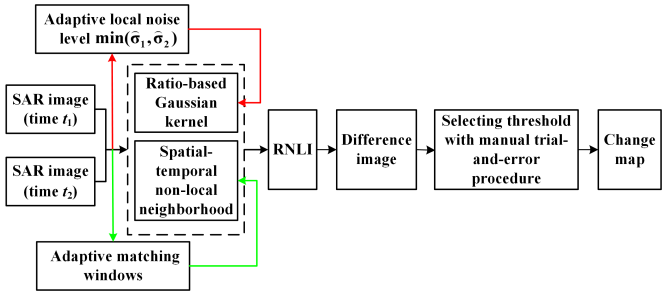


Fig. 1. General scheme of the proposed approach.

resolution of image [26] and video [27], and digital elevation model generation [28]. Considering the advantage that the information extraction is not limited to a local neighborhood, the nonlocal theory has great potential in change detection of SAR images. Nevertheless, NLM was designed to suppress the influence of additive noise in the image. Thus the change detection performance of NLM-based method is poor if NLM was directly used in change detection of SAR images with multiplicative noise.

To this end, an approach using ratio-based nonlocal information (RNLI) is proposed for change detection of SAR images in this study. The general scheme of the proposed RNLI is shown in Fig. 1. First the RNLI is extracted from a spatial–temporal nonlocal neighborhood, where the proposed ratio-based Gaussian kernel function is used for characterizing the similarity of two pixels in nonlocal neighborhood. The change detection performance of the proposed method is mainly affected by the noise level and the matching window size. If the noise level and the matching window size of RNLI were empirically determined, the change detection performance of RNLI may be worse than that of other methods based on local neighborhood information. To overcome this problem, a method for adaptively estimating the noise level is proposed in this study. In addition, the method for selecting the best filtering window size [29] is employed to determine the matching window size adaptively. Then the difference image is obtained by using RNLI which is constructed based on the ratio operator. Finally, the change map is obtained by segmenting the difference image with a threshold selected with manual trial-and-error procedure (MTEP). The method proposed in this study can reasonably and effectively use the spatial–temporal neighborhood information of multitemporal SAR images, reduce the dependence of change detection method on user experience, and improve the reliability of change detection result.

This article is organized into five sections. Section II presents the proposed method. Experimental datasets and results are shown in Section III. The discussion of the experimental results and the proposed method is shown in Section IV. Finally, conclusions are drawn in Section V.

II. Methods

A. NLM

The neighborhood information of a pixel is usually extracted from a square window centered on the pixel. The mean of a neighborhood is a simple, effective and widely used neighborhood information. However, the pixels in a local window are usually not enough for estimating a reliable mean. Hence, NLM was proposed by Buades *et al.* [21], which is not limited to a local window of an image.

Generally, the NLM at position x in an image is obtained by the following equation [21]:

$$u_{\mathrm{NLM}}(x) = \sum_{i \in \Omega_x} w(i,x) I(i) \tag{1}$$

where $u_{\mathrm{NLM}}(x)$ is the value of NLM at position x, $w(i,x)$ is the weight that characterizes the similarity between the pixels at positions i and x. And $w(i,x)$ satisfies the general characteristics of the weight: $0 \le w(i,x) \le 1$ and $\sum_{i \in \Omega_x} w(i,x) = 1$. $I(i)$ represents the intensity value of the pixel at position i, Ω_x represents the neighborhood of the search window corresponding the pixel at position x. $w(i,x)$ is calculated by

$$w(i,x) = \frac{1}{Z(x)} \exp\left(-\frac{||\mathbf{V}(i) - \mathbf{V}(x)||_{2,a}^2}{h^2}\right) \tag{2}$$

where $\mathbf{V}(i)$ and $\mathbf{V}(x)$, respectively, represent the vectors composed of pixels in the neighborhood at positions i and x, and the size of the neighborhood is equal to the matching window; h is the noise suppression parameter, $a > 0$ is the standard deviation of a Gaussian kernel; $Z(x) = \sum_{i \in \Omega_x} \exp(-(||\mathbf{V}(i) - \mathbf{V}(x)||_{2,a}^2 / h^2))$ is the normalizing parameter.

B. Ratio-Based Gaussian Kernel

The Gaussian kernel function has been widely used in image processing area, which could be expressed as

$$G(z) = \exp\left(-\frac{(z-u_z)^2}{2\sigma_z^2}\right) \tag{3}$$

where z is a variable of Gaussian kernel function, u_z is the mean, σ_z is the standard deviation. Equation (2), used to compute the similarity weight in NLM, is similar to the Gaussian kernel function shown in (3). And $h = 10\sigma$ was empirically determined in the first literature on NLM [21]. It can be seen that $h = \sqrt{2}\sigma$ is theoretically reasonable from a mathematical perspective (i.e., the Gaussian kernel function).

For the pixel-based change detection method of SAR images, it has been proven that the ratio operator is better than the subtraction operator as ratio operator can reduce the influence of radiometric calibration error [14]. Actually, the advantage of the ratio operator is benefit from the multiplicative speckle noise feature of SAR images. Inspired by the above, the ratio-based Gaussian kernel function is proposed in this study, which has two variables y_1 and y_2, and is different from the traditional Gaussian kernel function with one variable. The ratio-based Gaussian kernel function can be expressed as

$$G(y_1, y_2) = \exp\left(-\frac{(1 - \min(y_1, y_2)/\max(y_1, y_2))^2}{2\sigma^2}\right) \tag{4}$$

where σ is the standard deviation and is also called as noise level in this study, the $\min(\cdot)$ and $\max(\cdot)$ functions could

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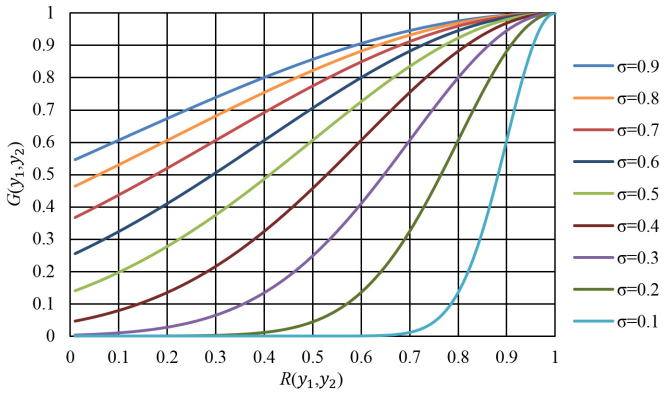


Fig. 2. Curve of $G(y_1, y_2)$ versus $R(y_1, y_2)$.

ensure that $G(y_1, y_2)$ is equal to $G(y_2, y_1)$. And this is a very important property because the similarity of y_1 and y_2 does not change by changing their order. Let $R(y_1, y_2)$ equal to $\min(y_1, y_2)/\max(y_1, y_2)$, then $G(y_1, y_2)$ is equal to $\exp\left(-(1 - R(y_1, y_2))^2/2\sigma^2\right)$. When the σ is 0.1–0.9, the curve of $G(y_1, y_2)$ versus $R(y_1, y_2)$ is shown in Fig. 2. It can be seen that the smaller the $R(y_1, y_2)$, the smaller the $G(y_1, y_2)$, and the smaller similarity of y_1 and y_2.

C. RNLI

Multitemporal SAR images are used for the special change detection problem. If both spatial and temporal nonlocal neighborhoods can be used simultaneously, it is likely to extract more reliable nonlocal information, thereby obtaining more accurate change detection results. Therefore, the "nonlocal" in RNLI contains not only spatial nonlocal neighborhood but also temporal nonlocal neighborhood, which is different from the traditional NLM only uses the spatial nonlocal neighborhood.

The change detection method based on RNLI mainly includes the following three steps. First, confirming the value of σ in (4). Second, confirming the matching window and the search window of RNLI. Third, the following equation is used to calculate the difference image based on RNLI:

$$D_{\mathrm{RNLI}}(x) = 1 - \frac{\min\{u_{\mathrm{RNLI1}}(x), u_{\mathrm{RNLI2}}(x)\}}{\max\{u_{\mathrm{RNLI1}}(x), u_{\mathrm{RNLI2}}(x)\}} \tag{5}$$

where D_{RNLI} represents the difference image obtained with RNLI; $D_{\mathrm{RNLI}}(x)$ represents the pixel value at position x in D_{RNLI}; $u_{\mathrm{RNLI1}}(x)$ represents the RNLI at position x in image I_1; $u_{\mathrm{RNLI2}}(x)$ represents the RNLI at position x in image I_2. The unchanged pixel in D_{RNLI} has a small intensity value while the changed pixel has a large intensity value.

In this study, $u_{\mathrm{RNLI1}}(x)$ is calculated by the following equation (similarly, $u_{\mathrm{RNLI2}}(x)$ can be obtained):

$$u_{\mathrm{RNLI1}}(x) = \sum_{i \in \Omega_x''} w''(t, i, 1, x) I(t, i), \quad t = 1, 2 \tag{6}$$

where $I(t, i)$ is the intensity value of the pixel at position i in image I_t; Ω_x'' represents the spatial–temporal nonlocal neighborhood, which is composed of the search windows corresponding to the pixels at position x in images I_1 and I_2; $w''(t, i, 1, x)$ is the weight characterizing the similarity between the pixel at position i in image I_t and the pixel at position x in image I_1. And $w''(t, i, 1, x)$ is obtained by the following equation, which is based on the proposed ratio-based Gaussian kernel function:

$$w''(t, i, 1, x) = \frac{1}{Z''(1, x)} \exp\left(-\frac{||1 - \mathbf{R}||_a^2}{2\sigma^2}\right), \quad t = 1, 2 \tag{7}$$

where $a > 0$ is the standard deviation of a Gaussian kernel; $Z''(1, x) = \sum_{i \in \Omega_x''} \exp(-(||1 - \mathbf{R}||_a^2/2\sigma^2))$, $t = 1, 2$ is the normalizing parameter, $\mathbf{R} = \min(\mathbf{V}(t, i), \mathbf{V}(1, x))/\max(\mathbf{V}(t, i), \mathbf{V}(1, x))$ is the ratio vector between $\mathbf{V}(t, i)$ and $\mathbf{V}(1, x)$. $\mathbf{V}(t, i)$ is the vector composed of the pixels in the neighborhood at positions i in image I_t, and the size of the neighborhood is equal to the matching window; $\mathbf{V}(1, x)$ is the vector composed of pixels in the neighborhood at positions x in image I_1, and the size of the neighborhood equates the matching window.

The proposed change detection method based on RNLI contains three parameters: the noisy level σ, the matching window size, and the search window size. Among them, the σ and the matching window size are two mainly factors affecting the change detection performance of the proposed method. The σ and the matching window size could be empirically determined like the traditional NLM; however, it will reduce the accuracy and reliability of RNLI-based change detection method. To overcome this problem, a method for adaptive estimating the local noise σ is proposed in this study. In addition, the adaptive selection method of the optimal window, usually used in the adaptive filtering method [29], is employed to select the matching window size in this study. Generally, the search window is a square window and is larger than the matching window. Each pixel inside the search window participates in the calculation for the nonlocal information of the center pixel. The search window with a size 21×21 is recommended by Buades *et al.* [21], [30] and is also adopted in this study.

D. Adaptive Noise Level

The σ of the ratio-based Gaussian kernel function is also named as noise level, which refers to the standard deviation of the noise and is one of the main factors affecting the detail retention of the changed area. If the noise level of an image can be adaptively estimated, the automation and accuracy of change detection can be improved. Given the fact that the multiplicative speckle noise is an inherent feature of SAR image, the intensity value I of an SAR image can be expressed as

$$I = I_0(1 + \delta) \tag{8}$$

where I_0 is the real intensity value, δ is the noise with zero mean. For an SAR image, the mean of a homogeneous neighborhood N_h with the real intensity I_0' can be computed by

$$u_h = \frac{1}{n_h} \sum_{x \in N_h} I_0'(1 + \delta_x) = \frac{1}{n_h} I_0' \left(n_h + \sum_{x \in N_h} \delta_x\right) \tag{9}$$

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where $x \in N_h$ means that the pixel at position x is located in N_h, δ_x is the noise value of the pixel at position x, n_h is the number of pixels included in N_h. $u_h = I'_0$ due to the mean of noise is zero. It can be seen that the mean of a homogeneous neighborhood has no relationship with noise. Therefore, the noise of the pixel at position x can be estimated by

$$\widehat{\delta}_x = \frac{I'}{u_h} - 1 \tag{10}$$

where $I' = I'_0(1 + \delta_x)$ is the intensity value of the pixel at x. It can be seen that the higher the homogeneity of a neighborhood, the closer the $\widehat{\delta}_x$ to the real noise δ_x. Therefore, the noise level can be well estimated based on the homogeneous area. And a method for adaptively estimating the noise level from an SAR image is proposed in this study.

The heterogeneity ∂, mathematically defined as the coefficient of variation, was usually used for selecting the homogeneous window from an SAR image [29]. In this study, ∂ is used to get the homogeneous neighborhood and is also employed in Section II-E for selecting the matching window. The ∂ of a homogeneous area is smaller than the ∂ of a heterogeneous area, thus the homogeneous area is characterized by a small ∂. The selection method of homogeneous neighborhood will be introduced in detail in Section II-E.

Hereafter, the noise variance for the neighborhood of the pixel at position x in the SAR image is unbiased estimated by

$$\widehat{\sigma}_R^2(x) = \frac{1}{n-1} \sum_{x_0 \in \Omega_x} \left| \frac{I(x_0)}{u_x} - 1 \right|^2 \tag{11}$$

where $\widehat{\sigma}_R(x)$ is the estimated noise level, Ω_x is the homogeneous neighborhood of the pixel at position x in the SAR image, n is the number of pixels included in Ω_x, u_x is the mean of Ω_x, $I(x_0)$ is the intensity value of the pixel at position x_0 in Ω_x. In order to improve the reliability of $\widehat{\sigma}_R(x)$, the recalculation of $\widehat{\sigma}_R(x)$ that only uses the pixels meet the requirement $|(I(x_0)/u_x) - 1| < 3\widehat{\sigma}_R(x)$ is recommended, which can reduce the estimation bias caused by a small number of nonhomogeneous pixels.

In the search window of RNLI, two pixels are more likely to belong to a homogeneous area when their ratio difference is less than the estimated noise level. As we all know, the probability that a Gaussian random variable is less than two times the standard deviation is 95.45%. And the probability that a Gaussian random variable is less than three times the standard deviation is 99.73%. Therefore, the estimated noise level divided by 2 or 3 is recommended to use in RNLI. Generally, there may exist a small difference between the estimated noise levels at the same position in multitemporal SAR images. To ensure that the RNLI values of the multitemporal SAR images are comparable, it is better to use the same noise level at the same position in the multitemporal images. To avoid the loss of image information caused by over-filtering, the minimum value of the local noise level at the same position x in the multitemporal images is used in this study.

E. Adaptive Matching Window

A homogeneous window composed of similar pixels is considered to be the optimal matching window. This is because the similarity weight calculated by (7) (shown in Section II-C) is more reliable when there is a high similarity between the center pixel and other pixels in the matching window. The heterogeneity and the homogeneity are two opposite measures, thus the ∂ of a neighborhood can be used to describe the homogeneity. The method reported in [29] is employed to select the matching window adaptively in this study. This method enables RNLI to use a large matching window in the homogeneous area and a small matching window in the heterogeneous area such as the edge of the changed area.

The optimal matching window for a given center pixel can hardly be known, however, its value range $[N_{\min}, N_{\max}]$ can be determined. For a pixel in an SAR image, the ideal matching window with $\partial = 0$ usually does not exist, which is limited by the noise of SAR images. If the ∂ of a window is under a threshold, it can be regarded as an optimal matching window. In this study, the heterogeneity threshold is automatically determined by using the maximum entropy threshold method [31]. There may be significant deviations when using small samples to estimate neighborhood information [32], thus the ∂ calculated from a large window is more reliable than a small one. Therefore, this study uses a window search strategy from large windows to small windows to determine the optimal matching window for the center pixel. And the algorithm for selecting the matching window adaptively is shown in Algorithm 1.

Algorithm 1 Adaptive matching window selection

Begin

(a) Set the minimum and maximum values of the matching window: $N_{\min}$ and $N_{\max}$.

(b) For a window size$N \in [N_{\min}, N_{\max}]$, calculate the corresponding heterogeneity threshold ∂_{TN} based on the maximum entropy threshold method.

(c) Set the current matching window $N = N_{\max}$ for the center pixel.

(d) Calculate the heterogeneity ∂_N of the current matching window, go to Step (e) if $\partial_N \geq \partial_{\mathrm{TN}}$; otherwise go to Step (f).

(e) Go to Step (f) if $N = N_{\min}$; otherwise, set $N = N-2$ and go to Step (d).

(f) Save the optimal matching window N for the current center pixel.

(g) If the current pixel is not the last pixel, move to the next pixel, and continue from Step (c); otherwise, the adaptive matching window of the entire image has been calculated, go to End.

End

The method to select the matching size has two input parameters: the minimum matching window $N_{\min}$ and the maximum matching window $N_{\max}$. Generally, when $N_{\min}$ is set to 3×3, RNLI has a good ability to retain the edge details of the changed area. However, considering that there may be

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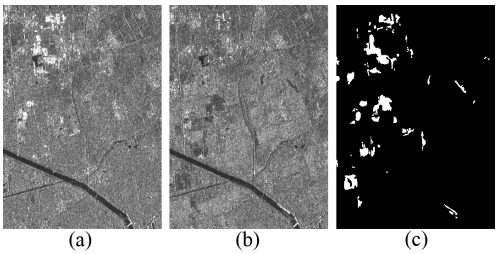


Fig. 3. Multitemporal images in Peixian. (a) Image acquired on June 13, 2017. (b) Image acquired on June 25, 2017. (c) Map of changed areas (ground truth) used as the reference in the experiments.

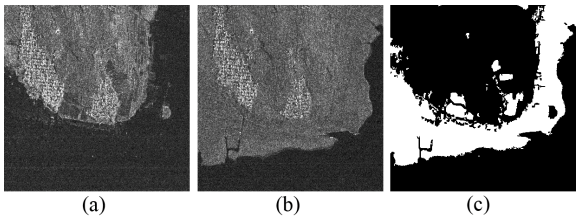


Fig. 4. Multitemporal images in Handan. (a) Image acquired in May, 2015. (b) Image acquired in November, 2015. (c) Map of changed areas (ground truth) used as the reference in the experiments.

significant deviations when small samples are used to estimate neighborhood information [32], and RNLI extracts the local neighborhood information more times than the change detection methods based on local neighborhood information (such as the classic MR). Thus a minimum matching window size of 5×5 is recommended for RNLI. The calculation amount of the adaptive selection algorithm for matching window increases with the increase of $N_{\max}$. The suitable range of $N_{\max}$ is 7×7–15×15, of which 11×11 is sufficient for most images [29].

III. Experimental Datasets and Results

Two real datasets and two simulated datasets are used in this study to test the effectiveness of the proposed RNLI method. Its results are compared with that of the LR [12], [16], second order log-cumulant (SOLC) [33], neighborhood-based principal component analysis (NPCA) [34], MR [20], NLM ratio (NLMR) [21], [35], cumulant-based Kullback–Leibler distance (CKLD) [20], and deep learning method based on convolutional neural network (DL-CNN) [36]. On the basis of Visual Studio2012 and OpenCV3.2, the experimental platform of this study is built with the C++ programing language.

A. Experimental Datasets

The real datasets used in this study are shown in Figs. 3 and 4. The first real dataset (with a size of 346×486) was acquired by the Sentinel-1 satellite on June 13 and 25, 2017, which is located in Peixian, China. The second dataset was acquired by Terra-SAR-X satellite in Handan, China. The size of it is 400×400, and the acquisition time was May 2015 and November 2015, respectively. The reference change maps shown in Figs. 3(c) and 4(c) were used to quantitatively analyze the performance of different change detection methods, which were defined manually according to an accurate and time-consuming visual inspection of the multitemporal SAR images.[1]

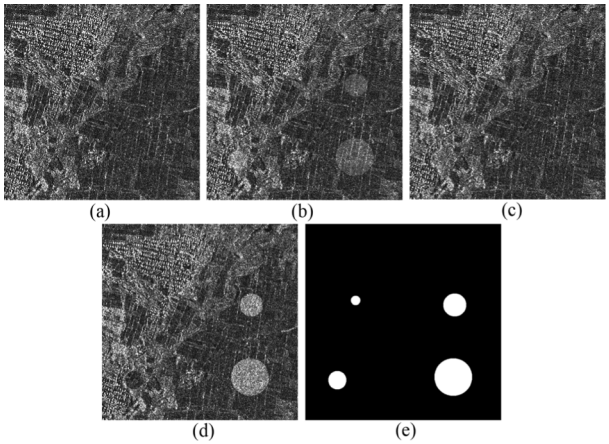


Fig. 5. Simulated dataset 1. (a) Original image. (b) Image with offset change. (c) Image with Gaussian change. (d) Image with deterministic change. (e) Map of changed areas (ground truth) used as the reference in the experiments.

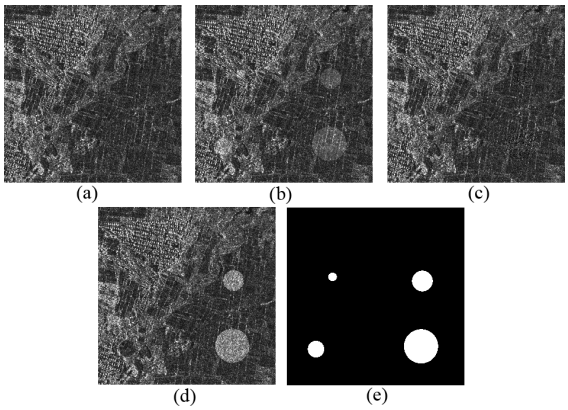


Fig. 6. Simulated dataset 2. (a) Original image. (b) Image with offset change. (c) Image with Gaussian change. (d) Image with deterministic change. (e) Map of changed areas (ground truth) used as the reference in the experiments.

The simulated datasets used in this study are shown in Figs. 5 and 6, which include three kinds of change. Fig. 5(a) (a Terra-SAR-X image with a size of 600×600) and its 2×2 multilook (Fig. 6(a) with a size of 300×300) were used to obtain the simulated datasets by using the change simulation method reported in [20]. The details of the three simulation changes are as follows: 1) the offset change is simulated by applying an offset value (i.e., a shift) to the image, which is useful to characterize the behavior of different detectors; 2) the Gaussian change is simulated by applying a zero-mean Gaussian additive noise to the image, which corresponds to a change in the state of the surface—field and vegetation; and 3) the deterministic change is simulated by pasting values copied from another area of the image, which occurs when there is a land use change, anthropic activities, etc. The comparison of the experimental results on simulated

[1] It is difficult to make quantitative comparisons of different change detection methods on dataset with large size because it is hard to obtain the reference change map from the dataset with large size.

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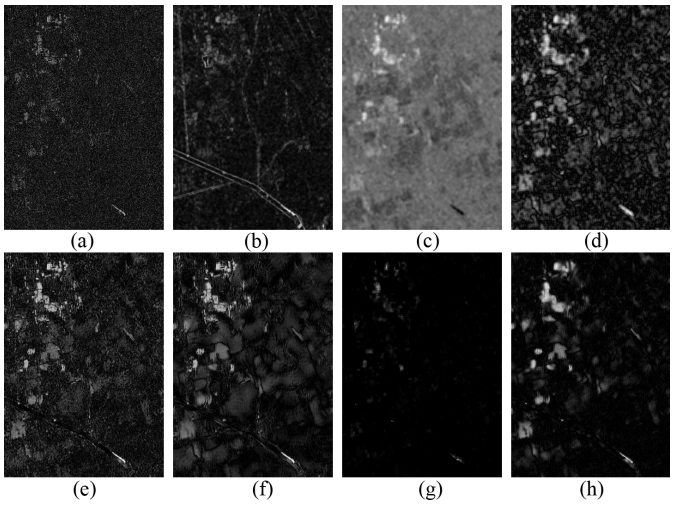


Fig. 7. Difference image obtained from the Peixian dataset with (a) LR, (b) SOLC 5 × 5, (c) NPCA 7 × 7, (d) MR 7 × 7, (e) NLMR 5 × 5, (f) NLMR 11 × 11, (g) CKLD 7 × 7, and (h) RNLI.

datasets is useful to further confirm the performance of the RNLI and will be shown in Section IV-C.

B. Difference Images of Peixian Dataset

First, the windows from 3 × 3 to 11 × 11 were used by the SOLC, NPCA, MR, and CKLD methods in the initial experiments. The results show that the 5 × 5 window for SOLC, and the 7 × 7 window for NPCA, MR, and CKLD are suitable for the Peixian dataset. The matching window settings that $N_{\min} = 5$ and $N_{\max} = 11$ are used by the RNLI. Thus, the matching windows 5 × 5 and 11 × 11 are used by NLMR for a better comparison with RNLI. The difference images generated from the Peixian dataset are shown in Fig. 7.[2] Comparing Fig. 7(a)–(h), it can be seen that the contrast of the changed area and the unchanged area in the difference image obtained by RNLI [Fig. 7(h)] is the highest and the edge contour of the changed area is the clearest. The contrast of the changed area and the unchanged area in the difference image obtained by the LR method [Fig. 7(a)] is the lowest. For the difference image obtained by SOLC [Fig. 7(b)], the contrast of the changed area and the unchanged area is low. In addition, some unchanged pixels located in the edge area have large intensity values, which makes it difficult to effectively identify the changed and unchanged pixels. Fig. 7(c) is the difference image obtained by NPCA, in which the changed area appears white, and the unchanged area appears gray or black. The difference image obtained by MR is shown in Fig. 7(d), in which the contrast of the changed area and the unchanged area is small. When the matching window is 5 × 5, the difference image obtained by NLMR is shown in Fig. 7(e), where the contrast of the changed area and the unchanged area is greater than that of SOLC, NPCA, and MR methods. When the matching window is 11 × 11, the difference image obtained by NLMR is shown in Fig. 7(f).

[2]It is worth noting that DL-CNN does not have a difference image because the last layer of its neural network model directly outputs change information.

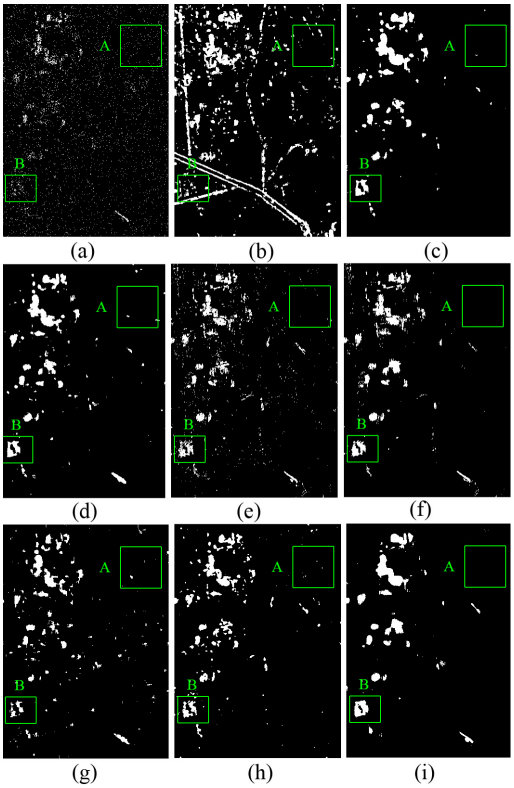


Fig. 8. Change maps obtained from the Peixian dataset by using (a) LR, (b) SOLC 5 × 5, (c) NPCA 7 × 7, (d) MR 7 × 7, (e) NLMR 5 × 5, (f) NLMR 11 × 11, (g) CKLD 7 × 7, (h) DL-CNN, and (i) RNLI.

Compared with Fig. 7(e), the unchanged area in Fig. 7(f) is smoother, but the edges of the changed area are blurred. Fig. 7(h) is the difference image obtained by the proposed RNLI. Fig. 7(g) is the difference image obtained by CKLD, in which the changed area appears white or gray, and the unchanged area appears black. Compared with other methods, RNLI can effectively extract the spatial–temporal information by using the ratio-based Gaussian kernel and nonlocal theory, so it performs well in terms of reducing the influence of speckle noise and retaining the edge details of the changed area.

C. Change Maps of Peixian Dataset

In this study, the change map is obtained by segmenting the difference image with a reasonable threshold (excluding DL-CNN which directly gets the change map). For the commonly used automatic threshold selection methods [1], [17], [37], the optimal threshold calculated under a certain assumption (such as the difference image obeys the Gaussian mixture distribution) is not always reasonable because the priori assumption is not always correct on different datasets [1], [38]. To avoid the negative effects caused by unreasonable thresholds, this study adopted the MTEP, which was widely used in the change detection literature [37]–[40]. The change maps resulting from the Peixian dataset are shown in Fig. 8. It can be seen that in the change map obtained by the LR method [Fig. 8(a)], there are some false alarm pixels in the unchanged area inside the green rectangle A. The false alarm pixels inside the green rectangle A in the change map

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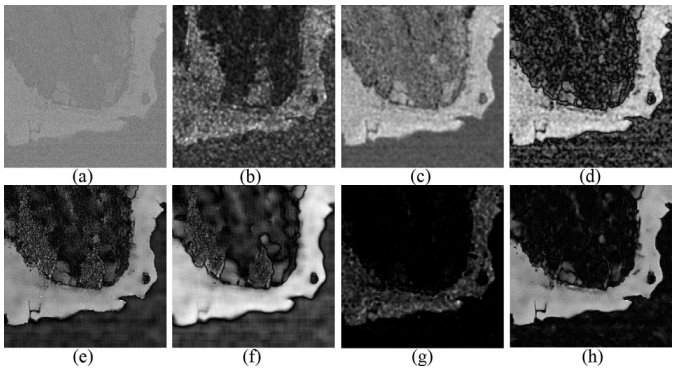


Fig. 9. Difference image obtained from the Handan dataset with (a) LR, (b) SOLC 7 × 7, (c) NPCA 7 × 7, (d) MR 7 × 7, (e) NLMR 5 × 5, (f) NLMR 11 × 11, (g) CKLD 7 × 7, and (h) RNLI.

obtained by MR [Fig. 8(d)] are fewer than those in Fig. 8(a), which reflect the advantage of using spatial neighborhood information for change detection. For Fig. 8(b) obtained by the SOLC, the unchanged pixels incorrectly detected as changed pixels are obviously larger than those corresponding to other methods, because the unchanged pixels in the heterogeneous area are detected as the changed pixels. In the green rectangle B, an unchanged area is surrounded by the changed pixels. Compared with Fig. 8(e), the unchanged area contained in rectangle B of Fig. 8(f) is better surrounded by the changed pixels, which shows the matching window 11 × 11 is more suitable than 5 × 5 in Peixian dataset. The changed area contained in rectangle B of Fig. 8(g) cannot enclose the unchanged area, because the proportion of the changed pixels in a 7 × 7 window is small when the window is located in the narrow changed area. In this situation, the estimated probability density function cannot well represent the center pixel. The change map obtained by RNLI is shown in Fig. 8(i), in which the edges of changed and unchanged areas in the rectangle B are clearer than those of other methods, which benefits from the combination of ratio-based Gaussian kernel and nonlocal theory.

D. Difference Images of Handan Dataset

The initial experiment for the Handan dataset is similar to that for the Peixian dataset. The windows from 3 × 3 to 11 × 11 were tested for the SOLC, NPCA, MR, and CKLD methods. And the results show that the window 7 × 7 is suitable for the Handan dataset. NLMR and RNLI use the same parameters as the Peixian dataset. Fig. 9 shows the difference images obtained by using different methods in the Handan dataset. The contrast of the changed area and the unchanged area in Fig. 9(a) is lower than that of other methods, because LR is a pixel-based changed detection method. The difference image obtained by the SOLC method [Fig. 9(b)] shows a clear block effect, and the intensity of some unchanged areas is similar to that of the changed area. This is because the intensity of the heterogeneous area is similar to that of the changed area in the difference image obtained by SOLC. Similar to the results of the Peixian dataset, the intensity of the unchanged area in the difference image [Fig. 9(g)] obtained by CKLD is smaller than that of other methods; however, the intensity of the changed area in Fig. 9(g) is also small. Compared with Fig. 9(h), the difference images obtained by NLMR with the matching windows 5 × 5 and 11 × 11 [Fig. 9(e) and (f)] have larger intensity in the unchanged area, which is because NLMR uses the same noise level for the entire image and is difficult to effectively suppress the negative effects of speckle noise.

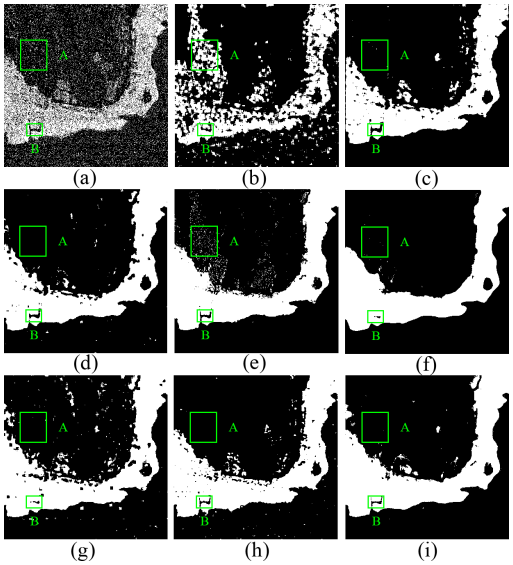


Fig. 10. Change maps acquired from the Handan dataset by using (a) LR, (b) SOLC 7 × 7, (c) NPCA 7 × 7, (d) MR 7 × 7, (e) NLMR 5 × 5, (f) NLMR 11 × 11, (g) CKLD 7 × 7, (h) DL-CNN, and (i) RNLI.

E. Change Maps of Handan Dataset

The change maps acquired from the Handan dataset are shown in Fig. 10, in which the green rectangle A is an unchanged area, and the green rectangle B contains both the changed and unchanged areas. Comparing Fig. 10, it can be seen that the change maps obtained by LR and SOLC are obviously worse than those obtained by other methods. In the change map acquired by LR, there are many discretely distributed false alarm pixels in the green rectangle A. The poor performance of SOLC is due to many unchanged areas in the green rectangle A are detected as changed areas by SOLC. For the methods using neighborhood information, in the green rectangle A, the SOLC and NLMR with a matching window of 5 × 5 have more false alarm pixels than other methods. Compared with RNLI, the green rectangle A in the change map obtained by NLMR with the matching window 5 × 5 has many false alarm pixels. In addition, the unchanged area in green rectangle B is detected as changed area when NLMR with the matching window 11 × 11 and CKLD are applied to get the change map. Considering the areas in rectangles A and B comprehensively, the change maps obtained by DL-CNN and RNLI are better than that obtained by the NLMR and CKLD methods.

IV. Discussion

A. Discussion on the Results of Peixian Dataset

Fig. 11 shows the receiver operating characteristic (ROC) [41], [42] curves of the difference images shown

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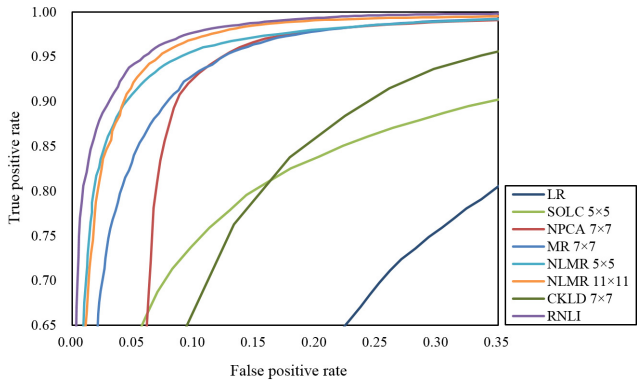


Fig. 11. ROC curves of the difference images obtained with different methods in the Peixian dataset.

TABLE I
AUCs OF DIFFERENT METHODS IN THE PEIXIAN DATASET

Method	LR	SOLC 5×5	NPCA 7×7	MR 7×7	NLMR 5×5	NLMR 11×11	CKLD 7×7	RNLI
AUC	0.794	0.893	0.932	0.968	0.978	0.981	0.906	0.988

in Fig. 7.[3] From the perspective of the ROC curve, a good change detection method has the characteristic that the ROC curve is close to the top left corner [12]. Comparing the eight ROC curves in Fig. 11, it can be seen that the difference image obtained by the RNLI method performs best because the ROC curve is closest to the top left corner. And the ROC curve corresponding to LR method is closest to the bottom right corner and has the worst performance. Comparing the ROC curves of the methods using spatial neighborhood information, the ROC curves corresponding to SOLC and CKLD are located at the bottom right, which show that SOLC and CKLD are worse than NPCA, MR, NLMR, and RNLI in change detection of Peixian dataset. The area under the ROC curves (AUCs) [43], [44] corresponding to the difference images in Fig. 7 are given in Table I. A good change detection method has a large AUC. Table I presents that the AUC corresponding to LR method is significantly smaller than those of SOLC, NPCA, MR, NLMR, CKLD, and RNLI. This is because LR method does not use the spatial neighborhood information of SAR images, and is greatly affected by speckle noise. In Table I, the RNLI method has the largest AUC, which also reflects the effectiveness of the proposed method.

The Kappa-threshold curves (KTCs) of the difference images obtained with different methods in Peixian dataset are shown in Fig. 12, which is used to compare the Kappa as a function of the varying threshold for each difference image. The comparison of KTCs has two advantages: 1) the maximum Kappa values of different methods can be compared together, the larger the Kappa, the better the performance of the method and 2) the curvature of the KTC around its maximum is a measure of threshold sensitivity, the lower the curvature, the lower the sensitivity of the method to the threshold. Comparing the KTCs shown in Fig. 12, it can be seen that the maximum Kappa values of LR and SOLC are obviously smaller than that of NPCA, MR, NLMR, CKLD, and RNLI. The maximum Kappa of RNLI is larger than that of other comparison methods. In addition, the curvature of the RNLI's KTC around its maximum is smaller than that of other methods, which also confirms the good change detection performance of RNLI.

[3]It is worth noting that there is no ROC curve and KTC corresponding to DL-CNN because DL-CNN does not generate a difference image. The change detection performance of the DL-CNN method can be compared through five quantitative analysis indicators of the change map.

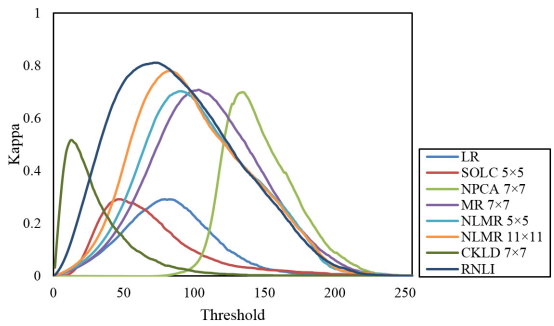


Fig. 12. KTCs of the difference images obtained with different methods in the Peixian dataset.

TABLE II
MISSED ALARMS, OVERALL ERROR, DETECTED CHANGES (IN NUMBER OF PIXELS), KAPPA, AND F1 SCORE OF THE CHANGE MAPS SHOWN IN FIG. 8

Method	Missed Alarms	Overall Error	Detected Changes	Kappa	F1 Score
LR	4837	8031	1854	0.291	0.316
SOLC 5×5	3023	15378	3668	0.292	0.331
NPCA 7×7	1980	3843	4711	0.698	0.710
MR 7×7	1871	3803	4820	0.705	0.717
NLMR 5×5	1908	3812	4783	0.703	0.715
NLMR 11×11	1741	2842	4950	0.768	0.777
CKLD 7×7	3331	5844	3360	0.517	0.535
DL-CNN	1945	3420	4746	0.725	0.735
RNLI	1512	2309	5179	0.811	0.818

The five quantitative analysis indicators of the change maps obtained from Peixian dataset are given in Table II. A good change map has small missed alarms and overall error, more detected changes, and large Kappa and F1 score. It can be seen from Table II that the LR method is worse than the other methods using neighborhood information, which shows the advantage of using neighborhood information in change detection of SAR images. In addition, DL-CNN performs worse than NLMR with the matching window 11×11, but its performance is better than that of SOLC, NPCA, MR, CKLD, and NLMR with the matching window 5×5. This shows that DL-CNN based on deep learning technology has certain competitiveness in change detection of SAR images. Compared with other methods, the change map obtained by RNLI has the smallest missed alarms and overall error, the largest detected changes, and the largest Kappa and F1 scores. Compared with the NLMR method, the good change detection performance of RNLI is beneficial from the use of the ratio-based Gaussian kernel and nonlocal theory.

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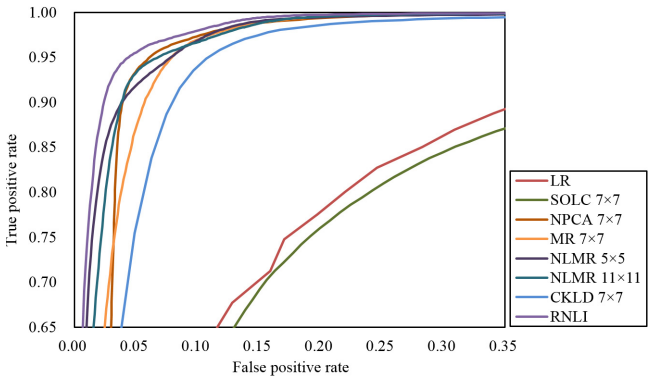


Fig. 13. ROC curves of the difference images obtained with different methods in the Handan dataset.

TABLE III
AUCs OF DIFFERENT METHODS IN THE HANDAN DATASET

Method	LR	SOLC 7×7	NPCA 7×7	MR 7×7	NLMR 5×5	NLMR 11×11	CKLD 7×7	RNLI
AUC	0.860	0.847	0.966	0.976	0.982	0.980	0.964	0.988

B. Discussion on the Results of Handan Dataset

The ROC curves of the difference images obtained with different methods in Handan dataset are shown in Fig. 13. The ROC curves corresponding to LR and SOLC are near the bottom right corner, which shows that the LR and SOLC methods perform worse than other methods. However, unlike the results in the Peixian dataset, the ROC curve corresponding to SOLC is at the bottom right of that corresponding to LR, which reflects that SOLC performs worse than LR on the Handan dataset. This is because both the changed area and the heterogeneous area have large intensity in the difference image obtained with SOLC. The ROC curve of CKLD is located at the top left of that corresponding to LR and SOLC methods, but at the bottom right of the ROC curves of the other methods. This shows that CKLD is better than LR and SOLC methods, but is worse than other comparison methods in Handan dataset. The ROC curve corresponding to RNLI is closest to the top left corner, which shows that RNLI has the best performance in change detection of Handan dataset. Compared with NLMR with the matching windows of 5×5 and 11×11, the experimental results also effectively reflect the advantages of using adaptive matching window, adaptive local noise level, and ratio-based Gaussian kernel in RNLI. The AUCs of the difference images in Fig. 9 are given in Table III. Similar to the situation reflected by the ROC curves, the LR and SOLC are not as good as the other methods because their AUCs are smaller than the AUCs of the other methods. In addition, RNLI with the largest AUC performs best in change detection of SAR images.

Fig. 14 shows the KTCs of the difference images shown in Fig. 9. It can be seen that the maximum Kappa values of LR and SOLC are smaller than those of other methods, which is consistent with the result of ROC curve. The maximum Kappa of RNLI is larger than that of other methods, which shows that RNLI has the best change detection performance.

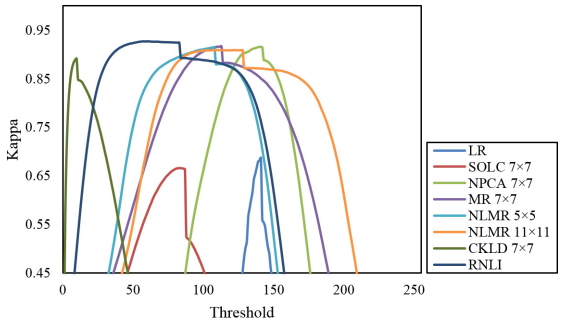


Fig. 14. KTCs of the difference images obtained with different methods in the Handan dataset.

TABLE IV
MISSED ALARMS, OVERALL ERROR, DETECTED CHANGES (IN NUMBER OF PIXELS), KAPPA, AND F1 SCORE OF THE CHANGE MAPS SHOWN IN FIG. 10

Method	Missed Alarms	Overall Error	Detected Changes	Kappa	F1 Score
LR	14747	29387	33012	0.687	0.691
SOLC 7×7	18538	31114	29221	0.518	0.652
NPCA 7×7	4906	7332	42853	0.921	0.921
MR 7×7	5166	7743	42593	0916	0.916
NLMR 5×5	5256	7980	42494	0.914	0.914
NLMR 11×11	5596	8846	42163	0.909	0.909
CKLD 7×7	6148	10072	41611	0.892	0.892
DL-CNN	5003	7501	42756	0.919	0.919
RNLI	3562	6939	44382	0.926	0.927

The curvature of the LR's KTC around its maximum is larger than that of the other methods. Furthermore, for NLMR and RNLI, the curvature of the KTCs around their maximum is smaller than that of other methods. This shows that the methods based on nonlocal theory are easily to obtain a change map with Kappa close to its maximum.

Similar to the Peixian dataset, the quantitative analysis indicators of the change maps obtained from the Handan dataset are calculated, which are given in Table IV. It can be seen from Table IV that the performance of LR is only better than SOLC, which is difficult to effectively identify the changed area. Similar to RNLI, SOLC also uses the spatial–temporal neighborhood information. However, the number of changed pixels detected by SOLC is smaller than that of other methods, the Kappa coefficient is slightly greater than 0.5, which is because SOLC only uses the local spatial–temporal information and has the disadvantage of identifying heterogeneous areas as changed areas. The NPCA, MR, CKLD, NLMR, and RNLI are significantly better than LR, because they use the neighborhood information and are less affected by speckle noise. Although the performance of NPCA in Handan dataset is only worse than RNLI, the NPCA's performance in Peixian dataset is only better than LR and SOLC, which shows that the change detection performance of NPCA is unstable. NLMR has different change detection performance when the different matching windows 5×5 and 11×11 are used, which shows that the change detection results will be uncertain if the matching window is set based on the user experience. If RNLI is not considered, DL-CNN is the only method

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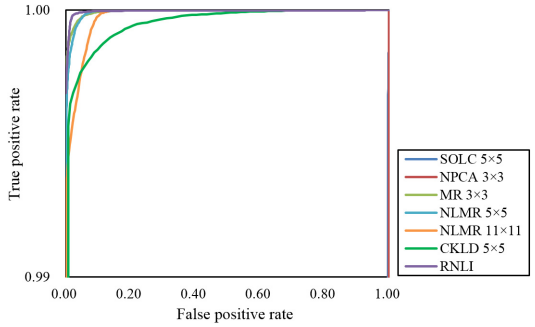


Fig. 15. ROC curves of different methods in the simulated dataset 1 (offset change).

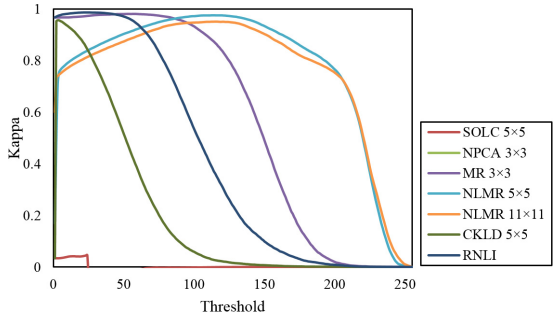


Fig. 16. KTCs of different methods in the simulated dataset 1 (offset change).

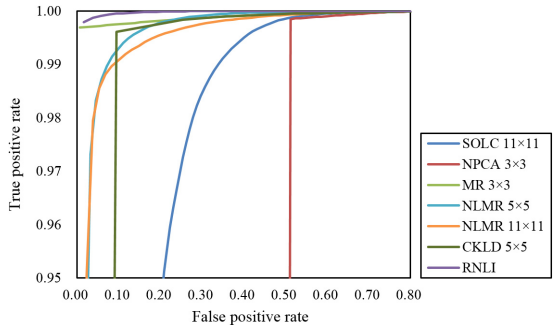


Fig. 17. ROC curves of different methods in the simulated dataset 1 (Gaussian change).

whose performance on two real datasets is always in the top three, which shows that the method based on deep learning technology has strong competitiveness in change detection of SAR images. Considering the five quantitative analysis indicators comprehensively, RNLI proposed in this study has the best performance in change detection, which shows the advantages of using nonlocal theory with ratio-based Gaussian kernel.

In summary, the five quantitative analysis indicators used in this study can well compare the performances of different methods on the same dataset, but it can also be seen that there are differences in the performances of the same method on the Peixian and Handan datasets. For example, the Kappa value of the same method on the Peixian dataset is smaller than that of the Handan dataset. The main reasons for this difference are as follows: 1) two datasets were obtained by Sentinel-1 and Terra-SAR-X satellites, respectively, and there are some differences in sensor parameters (wavelength, etc.) between the two satellites and 2) the difference between the change maps obtained by different methods is mainly in the edge area, for Peixian dataset, the percentage of changed pixels in the edge area to all changed pixels is larger than that in Handan dataset.

C. Discussion on the Results of Simulated Dataset 1

The neighborhood based methods are tested on the simulated datasets. The ROC curves of different methods in the simulated dataset 1 (offset change) are shown in Fig. 15. The huge difference among the ROC curves shows the performance difference of different change detection methods. The ROC curves of SOLC and NPCA are close to the right, which indicates that SOLC and NPCA are obviously worse than the other methods. The ROC curve of RNLI is closest to the top left corner, which presents that RNLI has the best performance in detecting offset change in simulated dataset 1. Fig. 16 shows the KTCs of different methods in the simulated dataset 1 (offset change). The maximum Kappa of CKLD is close to that of MR, NLMR, and RNLI, but the curvature of the CKLD's KTC around its maximum is larger than that of MR, NLMR, and RNLI. The maximum Kappa of RNLI is larger than that of other methods, which also confirms the good performance of the proposed method. There are differences between the KTCs of NLMR with different matching window sizes, which reflects the impact of the matching window size on change detection. In Fig. 16, the maximum Kappa of SOLC is significantly smaller than that of other methods. It is worth noting that we cannot see the KTC of NPCA from Fig. 16, which shows that NPCA is completely ineffective for detecting offset change. For NPCA, calculating the covariance of the data is a necessary step. However, after adding an offset to a dataset, the mean changes, the variance is the same as before, and the covariance is equal to the variance. In addition, the covariance between the same data is equal to the variance of the data. In summary, NPCA is not sensitive to offset change because the simulated offset change will not change the covariance of the data.

Fig. 17 shows the ROC curves of different methods in the simulated dataset 1 (Gaussian change). The difference among the ROC curves is smaller than that on Fig. 15. The performance of NPCA is worse than that of other methods. The good window size of SOLC is 11 $\times$ 11, which is obviously different from that of NPCA, MR, and CKLD. The ROC curve of RNLI is above that of other methods, which shows the good change detection performance of RNLI. The KTCs of different methods in the simulated dataset 1 (Gaussian change) are shown in Fig. 18. The maximum Kappa of NPCA is smaller than that of other methods. It can also be seen that the KTC of NPCA has the abrupt discontinuity, which appears in the experimental results of the simulated datasets 1 and 2 (Gaussian and deterministic changes). For the simulated datasets (Gaussian and deterministic changes), the covariance corresponding to the unchanged pixels is equal to the variance. As a result, in the difference images obtained by NPCA on the simulated datasets, the unchanged pixels have large and the same intensity value V_u. And the number of

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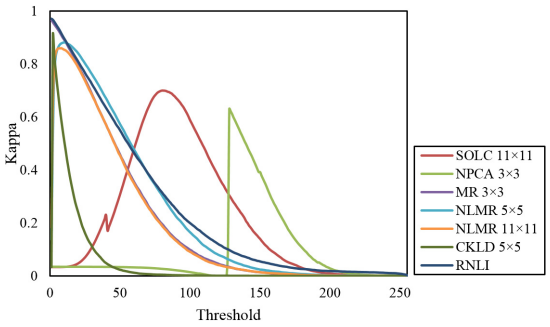


Fig. 18. KTCs of different methods in the simulated dataset 1 (Gaussian change).

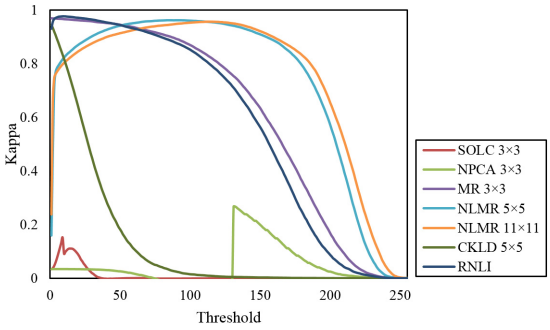


Fig. 20. KTCs of different methods in the simulated dataset 1 (deterministic change).

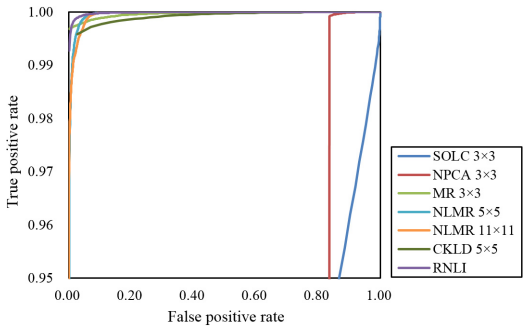


Fig. 19. ROC curves of different methods in the simulated dataset 1 (deterministic change).

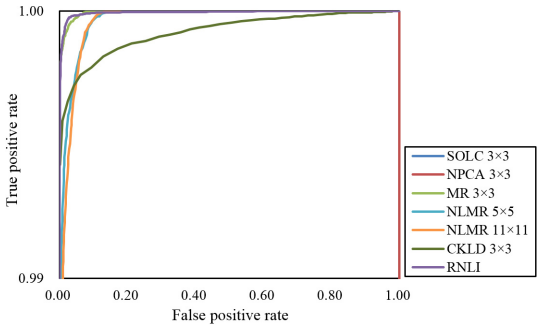


Fig. 21. ROC curves of different methods in the simulated dataset 2 (offset change).

false alarm pixels is particularly large before the threshold is greater than V_u. When the threshold is just greater than V_u, the false alarm pixel rapidly decreases, and the Kappa value rapidly increases, which manifests as the abrupt discontinuity in the KTC. Different from the results on offset change, the maximum Kappa of CKLD is obviously larger than that of SOLC, NPCA, and NLMR, which shows that the performance of CKLD on Gaussian change is better than that on offset change. Compared with the KTCs of MR and RNLI, the KTCs of SOLC, NPCA, NLMR, and CKLD have obvious peaks. However, the maximum Kappa of MR and RNLI is larger than that of other methods. Although the ROC curve of MR is below that of RNLI, there is only small difference between the maximum Kappa of MR and RNLI. This indicates that ROC curve and KTC are useful for comprehensive comparison of different methods.

The ROC curves of different methods in the simulated dataset 1 (deterministic change) are shown in Fig. 19. Different from the results on offset and Gaussian changes, the performance of NPCA is better than SOLC, which shows that the performance of SOLC and NPCA is different on different simulation changes. The ROC curve of RNLI is closest to the top left corner, which presents that RNLI performs better than the other comparison methods. Fig. 20 shows the KTCs of different methods in the simulated dataset 1 (deterministic change). It can be seen that the maximum Kappa of SOLC and NPCA is significantly smaller than that of other methods, which also shows the poor change detection performance of them. The maximum Kappa of RNLI is larger than that of other methods, which is similar to the result reflected by the ROC curve.

In summary, for the tested methods, there are differences among the ROC curves of different simulation changes, which also exists in the KTCs. This reflects that the change detection performance of different methods is different on different simulation changes. In particular, experimental results show that NPCA is completely ineffective for detecting offset change, and the NPCA's KTCs corresponding to Gaussian and deterministic changes have the abrupt discontinuities, both of which are limited by the principle of NPCA. Furthermore, compared with the simulated offset and deterministic changes, there is only small difference between the maximum Kappa of RNLI and MR in simulated Gaussian change. This is because RNLI is more suitable for SAR images with multiplicative noise. Moreover, RNLI is better than the other comparison methods to different degrees in different simulation changes.

D. Discussion on the Results of Simulated Dataset 2

Fig. 21 shows the ROC curves of different methods in the simulated dataset 2 (offset change). Similar to dataset 1, the ROC curves of SOLC and NPCA is close to the right. In addition, the ROC curve of RNLI is closest to the top left corner, which reflects that RNLI performs better than other methods. The KTCs of different methods in the simulated dataset 2 (offset change) are shown in Fig. 22. It is similar to the simulated dataset 1 that we cannot find the KTC of NPCA in Fig. 22. This situation is caused by the basic principle of NPCA, which has been discussed in Section IV-C.

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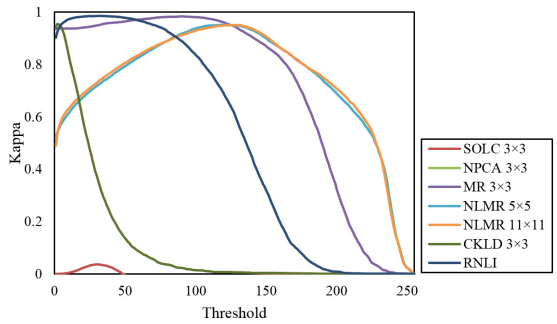


Fig. 22. KTCs of different methods in the simulated dataset 2 (offset change).

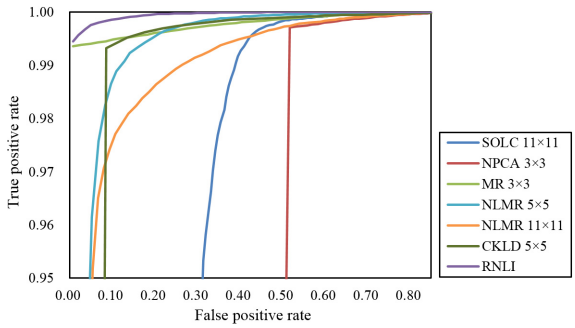


Fig. 23. ROC curves of different methods in the simulated dataset 2 (Gaussian change).

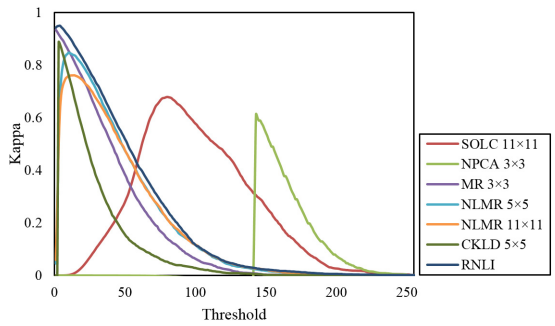


Fig. 24. KTCs of different methods in the simulated dataset 2 (Gaussian change).

The maximum Kappa of RNLI is larger than that of other methods, which shows the advantage of using nonlocal theory with ratio-based Gaussian kernel. In addition, the curvature of the RNLI's KTC around its maximum is small, which shows that the threshold with a large Kappa is easy to obtain.

The ROC curves of different methods in the simulated dataset 2 (Gaussian change) are presented in Fig. 23, which are similar to that of the simulated dataset 1 (Gaussian change). The ROC curve of RNLI is closest to the top left corner while the ROC curve of NPCA is close to the bottom right corner, which indicates that the performance of RNLI is the best while the performance of NPCA is poor. Fig. 24 shows the KTCs of different methods in the simulated dataset 2 (Gaussian change). Similar to the results of simulated dataset 1, the maximum Kappa of CKLD is larger than that of SOLC, NPCA, and NLMR, which shows that the performance of CKLD is similar on two simulated datasets (Gaussian change).

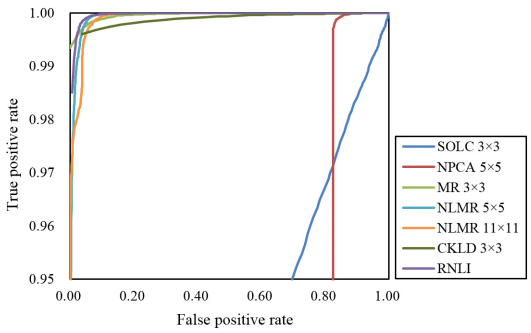


Fig. 25. ROC curves of different methods in the simulated dataset 2 (deterministic change).

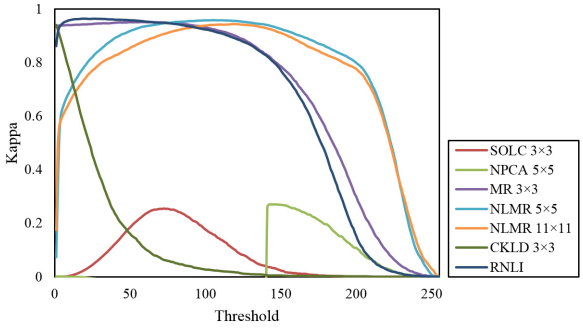


Fig. 26. KTCs of different methods in the simulated dataset 2 (deterministic change).

In addition, different from the KTCs shown in Fig. 18, the maximum Kappa of RNLI is obviously larger than that of MR. And the curvature of the RNLI's KTC around its maximum is smaller than that of MR. The curvature of the NPCA's KTC around its maximum is large, which reflects that it is hard to select a threshold with a large Kappa for NPCA. It is similar to the simulated dataset 1 that the KTC of NPCA has the abrupt discontinuity, which is caused by the basic principle of NPCA and has been discussed in Section IV-C.

Fig. 25 presents the ROC curves of different methods in the simulated dataset 2 (deterministic change). It can be seen that the ROC curves in Fig. 25 is similar to that in Fig. 19. The ROC curves of SOLC and NPCA are close to the lower right corner, which shows that the performance of SOLC and NPCA is worse than that of other methods. In addition, the ROC curve of RNLI is closest to the top left corner, which confirms that RNLI performs best in change detection of the simulated dataset 2 (deterministic change). The KTCs of different methods in the simulated dataset 2 (deterministic change) are shown in Fig. 26. The maximum Kappa of SOLC and NPCA is significantly smaller than that of other methods, which is similar to the results shown in Fig. 20. The maximum Kappa of RNLI is larger than that of other methods, and the curvature of the RNLI's KTC around its maximum is very small, which reflects the good performance of RNLI in detecting deterministic change in simulated dataset 2.

In summary, the main difference between the simulated dataset 1 and 2 is that the original image of simulated dataset 2 is obtained by applying 2×2 multilook to the original

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image of simulated dataset 1. For the same simulation change, the ROC curves of the tested methods in simulated dataset 2 are similar to that of simulated dataset 1, which also exists in the KTCs. This reflects that the change detection performance of different methods is similar on two simulated datasets. But there are also some small differences. For example, for the simulated deterministic change, the intersection between the KTCs of SOLC and NPCA is different in two simulated datasets. Moreover, for the simulated Gaussian change, the curvature of the RNLI's KTC around its maximum in simulated dataset 2 is smaller than that in simulated dataset 1, which shows that it is easy to select a threshold with a large Kappa for RNLI in simulated dataset 2.

E. Discussion on the Proposed Method

For change detection of multitemporal SAR images, an RNLI method was shown in this study, which has been compared with the state-of-the-art methods to confirm its effectiveness. RNLI was based on the nonlocal theory and the ratio-based Gaussian kernel. The nonlocal theory makes RNLI not limited to a local neighborhood when extracting information. For SAR image with multiplicative speckle noise, the proposed ratio-based Gaussian kernel enables RNLI to well characterize the similarity of the pixels in nonlocal neighborhood. In addition, given the fact that multitemporal SAR images are used for change detection problem, both spatial and temporal nonlocal neighborhoods were used in the proposed RNLI. The usage of spatial–temporal nonlocal neighborhood enables RNLI to well suppress the influence of speckle noise on change detection and obtain accurate change detection results.

The change detection performance of the proposed method is affected by two main parameters: the noise level and the matching window size.

1) The parameter σ of RNLI is used for controlling the noise suppression capability, which is also named as noise level. Generally, the noise level of an image is unknown and is set based on the user's experience. Suppose the noise level in an image is σ_0. If the noise level used in the RNLI is larger than σ_0, there will be over-filtering, which results in blurred edges of the changed area. If the noise level is smaller than σ_0, RNLI cannot effectively reduce the negative influence of speckle noise on change detection, thereby getting the unreliable change detection results. For the same dataset, there are differences among the change detection results when different users use different noise levels. This can be understood as the uncertainty of the change detection results caused by manually setting the noise level. In this study, this problem was overcome by proposing an adaptive estimating method of the noise level.
2) The matching window is a square window used to calculate the similarity between the center pixel and other pixels in the search window. The optimal matching windows for different pixels in an image are different. Selecting a small matching window is beneficial to retain edge detail information of the changed area but weakens the ability to reduce the false alarm pixels. By contrast, selecting a large matching window can better reduce the false alarm pixels, but may lose the edge details of the changed area. Therefore, when the user selects a fixed matching window for the entire image, it is necessary to make a compromise between the preservation of edge detail in the changed area and the reduction of the false alarm pixels of the change detection result. There are differences among the change detection results when different users select different matching windows for the same dataset, which can be understood as the uncertainty of the change detection results caused by manually setting the matching window. In this study, this problem was avoided by adaptively selecting the optimal matching window for each pixel, and the accuracy and automation of change detection can also be improved.

V. CONCLUSION

In this study, an RNLI method has been proposed for change detection in multitemporal SAR images, which combines the nonlocal theory with a novel ratio-based Gaussian kernel. On one hand, compared with the change detection methods based on local neighborhood, the proposed RNLI overcomes the limit of extracting information from a local neighborhood. On the other hand, compared with the NLMR, the proposed RNLI overcomes the limit of the traditional weight function designed for reducing the effect of additive noise. The proposed method can effectively use the spatial–temporal neighborhood information contained in multitemporal SAR images, thereby improving the accuracy of the change detection results.

The change detection performance of RNLI is mainly affected by two parameters: the noise level and the matching window size. To overcome the dependence on user experience, a method for adaptive estimating the local noise σ is proposed in this study. In addition, a method usually used in filtering methods is employed to adaptively select the matching window size. As a result, RNLI solves the problem that the change detection results have uncertainty because different users may adopt different empirical parameters (i.e., the fixed matching window and noise level), and has better usability in practical applications.

The experiments were conducted on two real datasets and two simulated datasets to prove the effectiveness of the proposed RNLI. Experimental results showed that the RNLI performed better than the other methods in change detection of SAR images. The main contributions of this study are as follows: 1) an RNLI method that does not rely on user experience was proposed for change detection of SAR images, which can better retain the edge information of the changed area while reducing the overall error of the change detection results; 2) a ratio-based Gaussian kernel function was proposed to characterize the similarity of SAR pixels with multiplicative speckle noise; and 3) a method was proposed to adaptively estimate the noise level of RNLI, which improved the practicality of the proposed RNLI.

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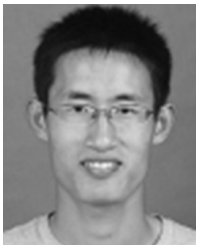
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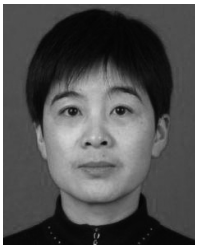
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